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Solvable Lie Algebras

1.1 Solvable Lie Algebras

Lemma 1.1.1

Suppose that I is an ideal of L . Then L/I is abelian if and only if I contains the derived algebra L' .

Definition 1.1.2

This lemma tells us that the derived algebra L' is the smallest ideal of L with an abelian quotient. By the same argument, the derived algebra L' itself has a smallest ideal whose quotient is abelian, namely the derived algebra of L' , which we denote $L^{(2)}$, and so on. We define the **derived series** of L to be the series with terms

$$L^{(1)} = L' \quad \text{and} \quad L^{(k)} = [L^{(k-1)}, L^{(k-1)}] \text{ for } k \geq 2.$$

Then $L \supseteq L^{(1)} \supseteq L^{(2)} \supseteq \dots$. As the product of ideals is an ideal, $L^{(k)}$ is an ideal of L (and not just an ideal of $L^{(k-1)}$).